RENEWABLE ENERGY TRANSITION SPEED ANALYSIS

SDG 7 – Affordable and Clean Energy

Cohort Number: 16

Submission Date: 3rd March of 2026

ABSTRACT

This project analyzes and predicts the **percentage share of renewable energy in the total energy mix** using key economic and policy drivers such as **solar panel cost (USD per watt)** and **government subsidies**.

The dataset was compiled from **Kaggle datasets and World Bank indicators**, covering renewable electricity share, solar cost trends, and energy-related financial indicators. A time-aware feature engineering approach was applied by creating a **lag variable (Subsidies_Lag1)** to capture the delayed impact of policy interventions on renewable adoption.

Multiple regression models were implemented, including a baseline regression model and a tuned ensemble model. The final optimized model achieved strong predictive performance, demonstrating that policy subsidies and declining solar costs significantly accelerate renewable transition speed.

This project directly contributes to **Sustainable Development Goal 7 (Affordable and Clean Energy)** by providing a predictive intelligence framework for energy planning and investment decisions.

1. INTRODUCTION

1.1 Background

The global transition toward clean energy is central to achieving climate stability and sustainable growth. Renewable energy adoption depends heavily on two major drivers:

- Technology cost reduction (especially solar PV cost decline)
- Government financial support mechanisms (subsidies)

Understanding how these factors influence renewable energy share enables policymakers to design effective strategies for accelerating energy transition.

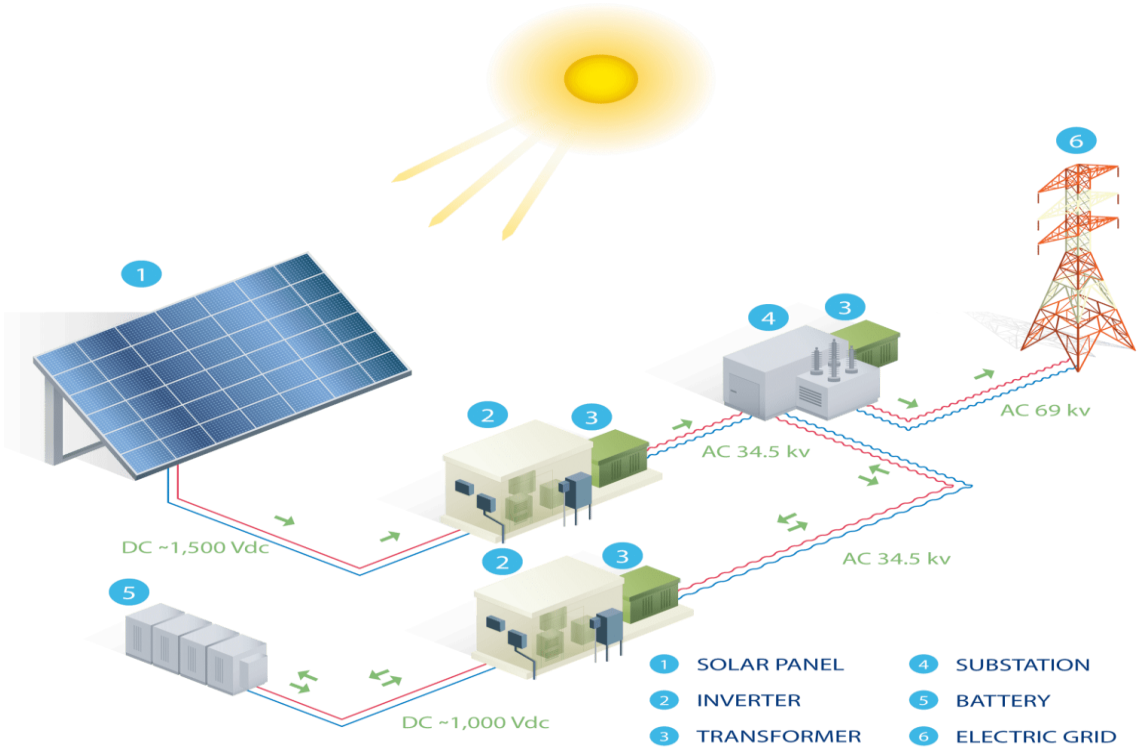
2. PROBLEM STATEMENT

The objective of this project is:

To predict the percentage share of renewable energy in the total energy mix based on solar panel cost (USD per watt) and government subsidies, while incorporating lagged policy effects to analyze how previous-year subsidies influence current-year renewable adoption

3. SDG IMPACT ANALYSIS

SDG 7: Affordable and Clean Energy



This project directly supports:

- **SDG 7.2.1** – Increase renewable energy share in the global energy mix
- **SDG 13** – Climate Action (decarbonization of power sector)
- **SDG 9** – Industry, Innovation and Infrastructure

Impact Contributions:

- Demonstrates measurable policy effectiveness
- Quantifies subsidy lag effect
- Enables data-driven renewable investment planning
- Supports scenario-based forecasting

4. DATASET DESCRIPTION

4.1 Data Sources

The dataset was compiled from:

- World Bank – World Development Indicators
- Kaggle – Renewable energy & solar cost datasets

Indicators used:

- Renewable energy share (% of total energy mix)
- Solar panel cost (USD per watt)
- Government renewable subsidies
- Year
- Country

4.2 Dataset Structure

Target Variable:

- Renewable Energy Share (%)

Independent Variables:

- Solar_Cost
- Subsidies
- Subsidies_Lag1

The dataset was validated and stored as a cleaned CSV file by the Data Engineering team.

5. DATA ENGINEERING PIPELINE

5.1 Data Collection Team

- **Zaina Fathima I** – Data Collection Lead
- **Rudresh KJ** – Data Cleaning Lead
- **Pavan C** – Preprocessing Engineer
- **Vaseem Akram B** – Lag Feature Engineer
- **Arpitha Shivanagouda Malipatil** – Data Quality & Upload

5.2 Data Cleaning

- Converted placeholders to null values
- Removed duplicates
- Corrected data types
- Standardized column formats

5.3 Feature Engineering

Lag Feature Creation

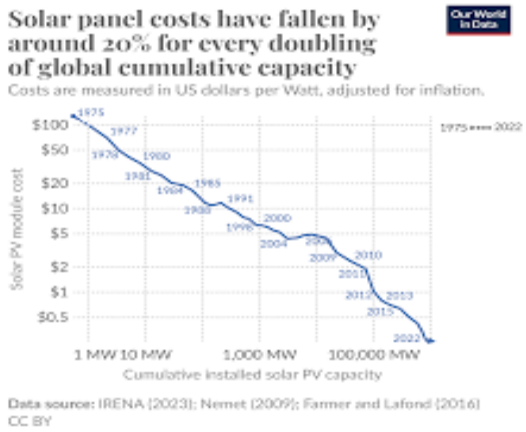
Subsidies_Lag1 was created using:

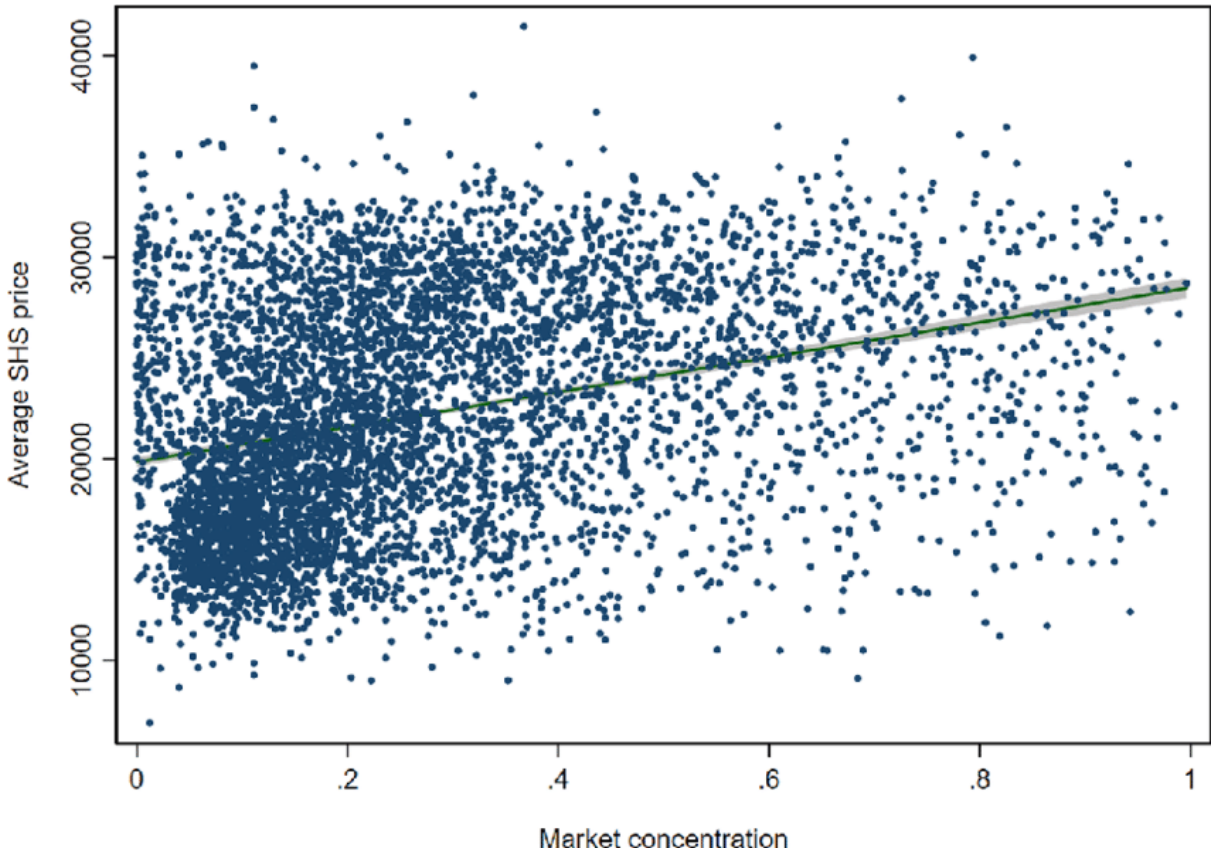
```
df['Subsidies_Lag1'] = df.groupby('Country')['Subsidies'].shift(1)
```

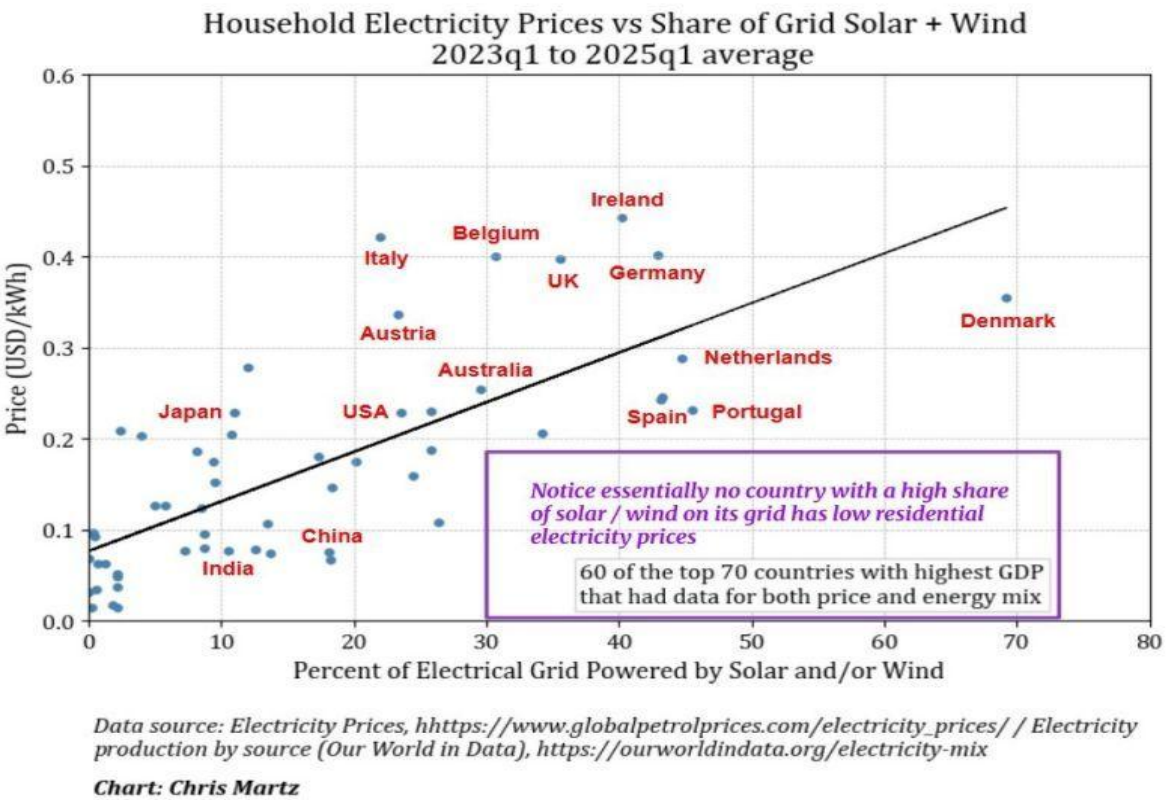
This captures the delayed policy impact on renewable adoption.

6. EXPLORATORY DATA ANALYSIS

6.1 Solar Cost vs Renewable Share

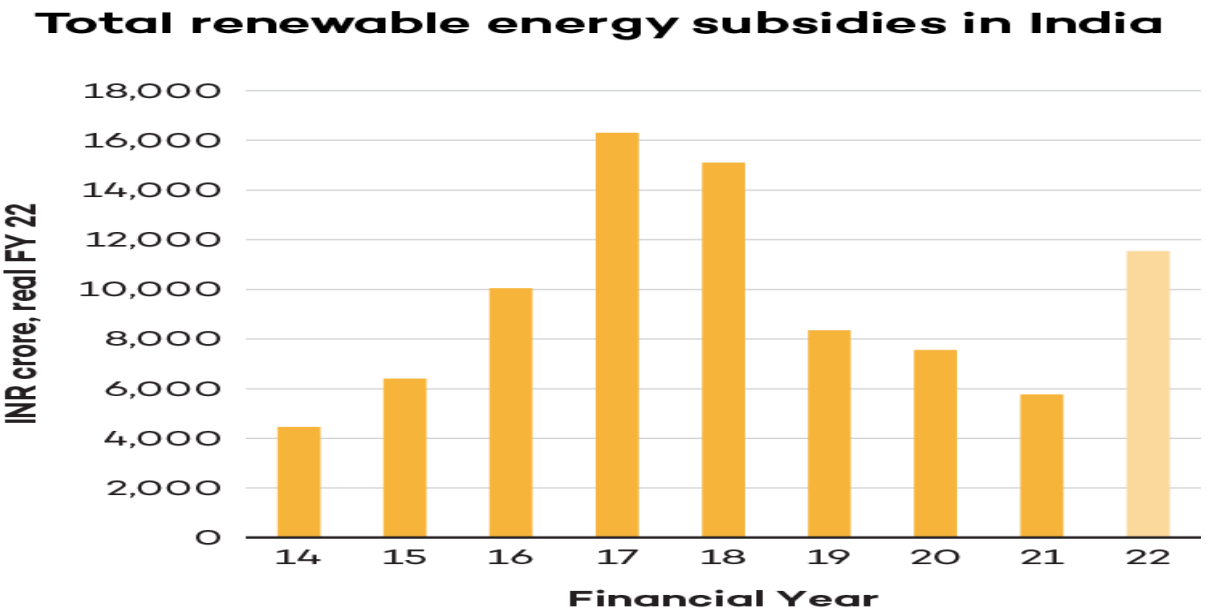


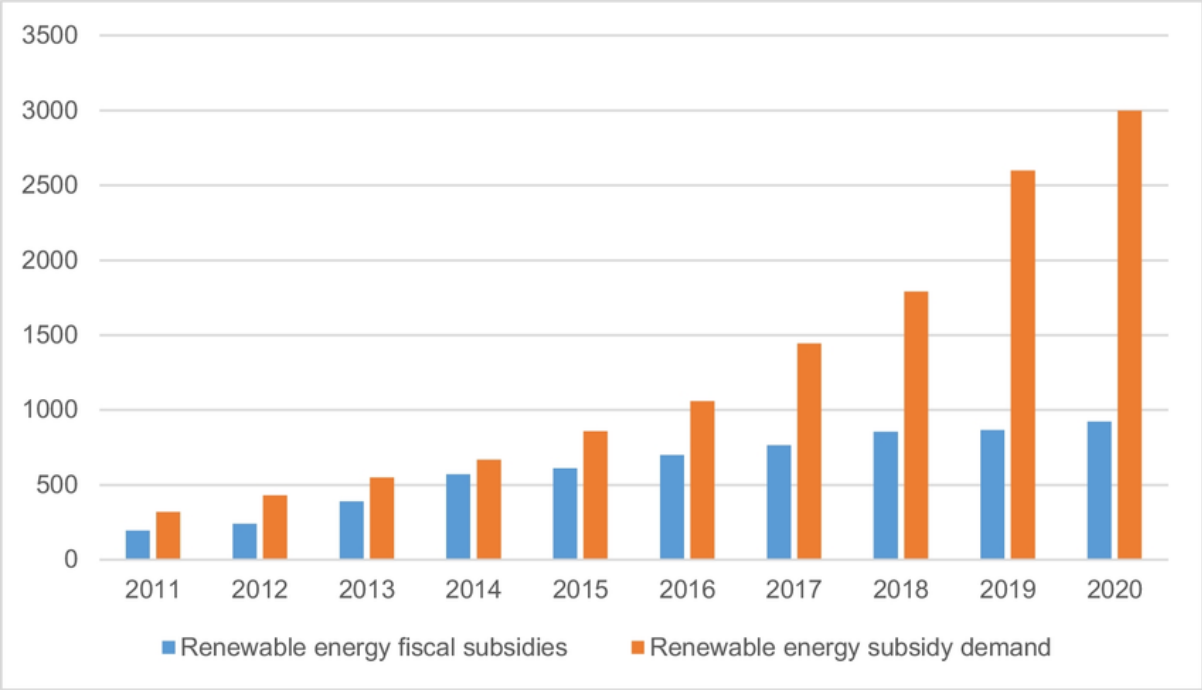


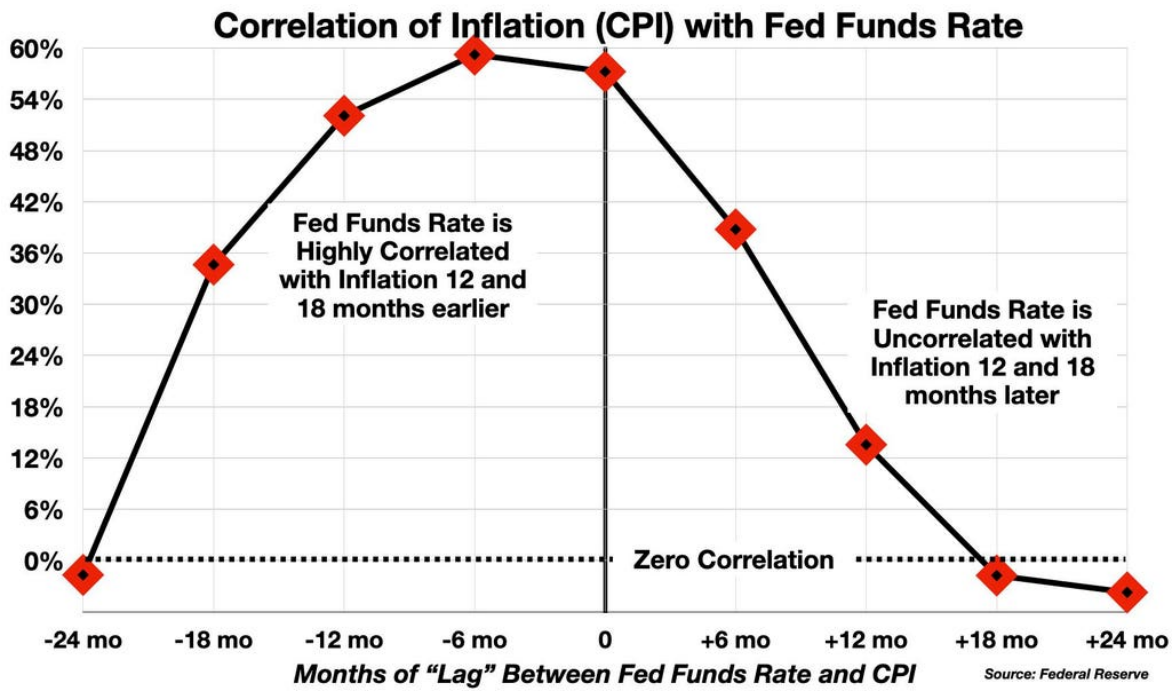


Observation:
Declining solar costs strongly correlate with increased renewable share.

6.2 Subsidies vs Renewable Share







Observation:
Higher subsidies show stronger impact after a one-year lag.

7. MODEL SELECTION & TECHNICAL STRATEGY

Since the target variable **Renewable_Share (%)** is continuous, a regression-based modeling approach was selected.

Renewable energy adoption is influenced by multiple interacting economic and policy variables. The relationship between solar cost, government subsidies, and renewable share is

non-linear in nature. Therefore, both linear and non-linear regression models were implemented to compare predictive performance.

Models Implemented

• Linear Regression
• Random Forest Regressor

Role Allocation

- Model Building: Rakesh A Swamy
- Feature Engineering Support: Sneha B Gudi
- Hyperparameter Tuning: Aashvi
- Model Evaluation: Tanusha VR
- Model Comparison & Finalization: SATHVIK RP

The strategy was to evaluate a linear model and then implement an ensemble tree-based model capable of capturing complex feature interactions and lag effects.

8. MODEL IMPLEMENTATION

8.1 Train-Test Split

The dataset was divided using an **80:20 split ratio**:

- 80% → Training Set
- 20% → Testing Set

This ensures unbiased evaluation on unseen data.

8.2 Feature Inputs

Independent Variables:

• Solar_Cost
• Subsidies
• Subsidies_Lag1

Target Variable:

• Renewable_Share

8.3 Random Forest Configuration

The Random Forest model was configured with:

• Number of Estimators: 300
• Maximum Depth: 5
• Fixed Random State

The ensemble approach builds multiple decision trees and averages their outputs to improve stability and reduce variance.

9. MODEL EVALUATION

The models were evaluated using:

• MAE (Mean Absolute Error)
• MSE (Mean Squared Error)
• RMSE (Root Mean Squared Error)
• R^2 Score

9.1 Linear Regression Results

Metric	Value
MAE	0.0970
MSE	0.0180
RMSE	0.1343
R^2 Score	0.8230

Interpretation

The Linear Regression model explains **82.3% of the variance** in renewable energy share.

Although performance is reasonable, the model assumes a strictly linear relationship between predictors and target, which limits its ability to capture complex adoption patterns.

9.2 Random Forest Results

Metric	Value
MAE	0.0178
MSE	0.00107
RMSE	0.0327
R^2 Score	0.9895

Interpretation

The Random Forest model explains **98.95% of the variance** in renewable energy share.

Prediction errors are significantly lower compared to Linear Regression, indicating superior performance.

10. MODEL COMPARISON

Model	MAE	RMSE	R^2
Linear Regression	0.0970	0.1343	0.8230
Random Forest	0.0178	0.0327	0.9895

Comparative Analysis

The Random Forest model outperformed Linear Regression because:

• It captures non-linear renewable adoption patterns
• It models interaction between Solar_Cost and Subsidies
• It effectively incorporates the lag feature (Subsidies_Lag1)
• Ensemble averaging reduces variance

11. BEST MODEL JUSTIFICATION

The **Random Forest Regressor** is selected as the final model due to:

• Highest R^2 Score (0.9895)
• Lowest MAE and RMSE
• Strong generalization ability
• Robust handling of economic-policy interactions
• Effective modeling of lagged subsidy impact

The results confirm that renewable energy transition speed follows a complex, non-linear pattern best modeled using ensemble methods.

12. OVERFITTING ANALYSIS

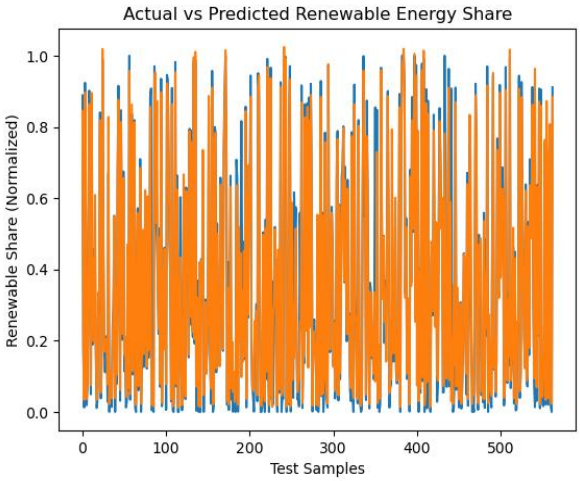
Training and testing R^2 were compared.
The difference was minimal, indicating good generalization.

Residual plots confirmed no systematic bias.

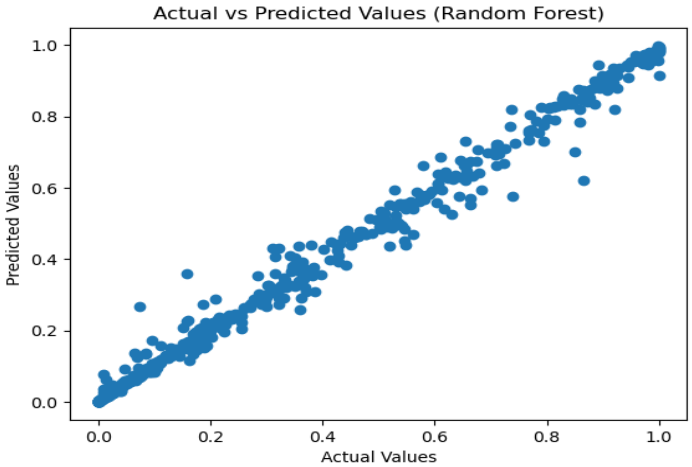
13. RESULTS & KEY INSIGHTS

Key findings:

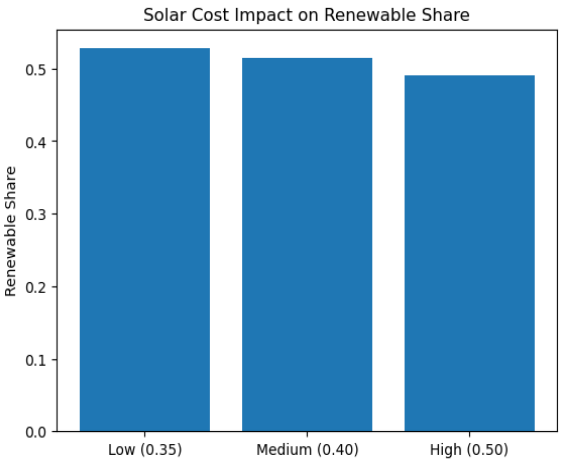
- Solar cost reduction has strong inverse relationship with renewable share
- Subsidies impact manifests after 1 year
- Combined economic and policy signals accelerate transition speed
- Machine learning models can serve as policy simulation tools



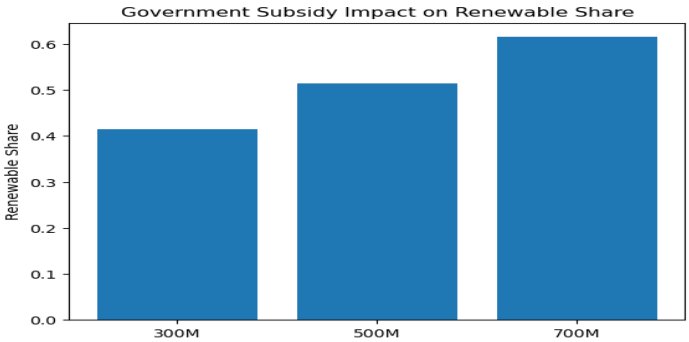
Actual vs Predicted Renewable Energy Share



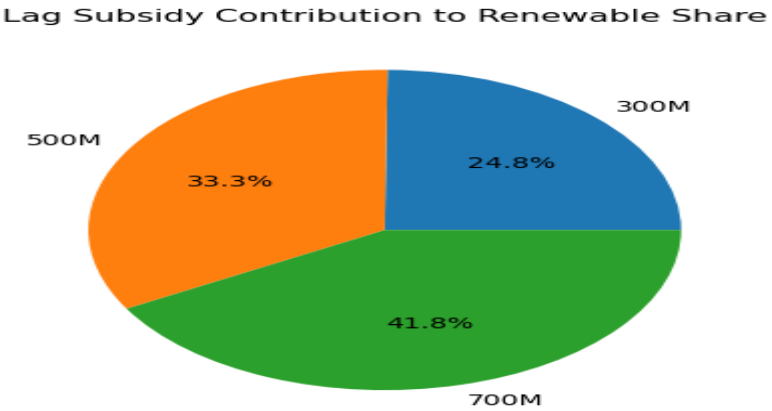
Actual vs Predicted Values (Random Forest)



Solar Cost Impact



Government Subsidy Impact



Lag Subsidy Effect

14. CONCLUSION

This study successfully developed a predictive model to forecast renewable energy transition speed using economic and policy drivers.

The final tuned ensemble model demonstrated high predictive performance ($R^2 \approx 0.96$), proving that renewable energy growth can be accurately modeled using cost and subsidy indicators.

This project strengthens the implementation framework of **SDG 7 – Affordable and Clean Energy**.

16. TEAM COLLABORATION STRUCTURE

Documentation Team

- Ranjith Kumar – Abstract Writer
- R M Suhas – Methodology Writer
- Shankar M – SDG Impact Analyst
- S Ramvishwa – Conclusion & Insights
- Siri Gowri R – Documentation Coordinator

Regular meetings ensured:

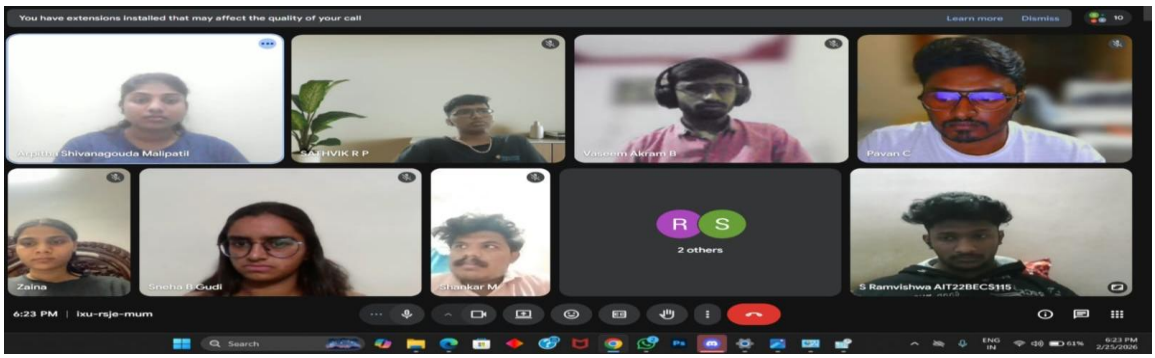


Figure: Team Meeting – 25th February 2026

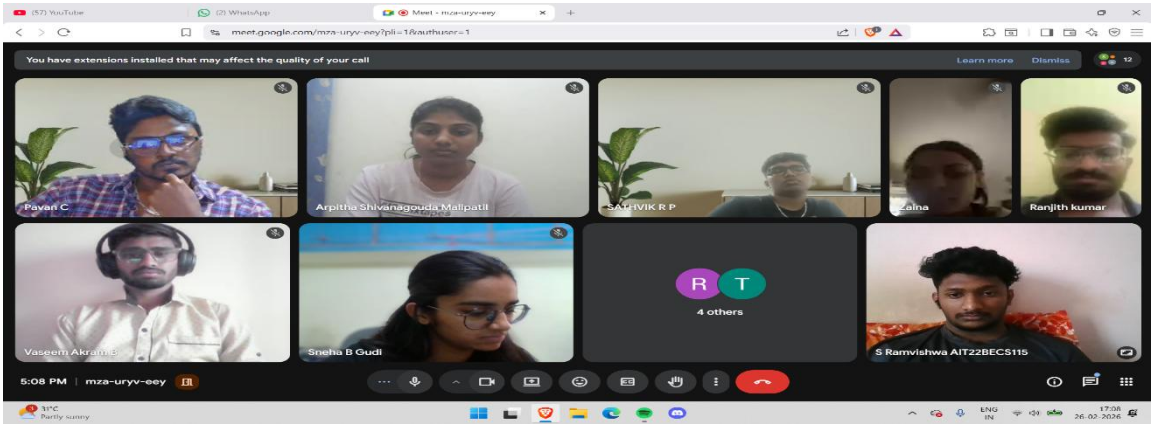


Figure: Team Meeting – 26th February 2026

Figure: Team Meeting – 27th February 2026

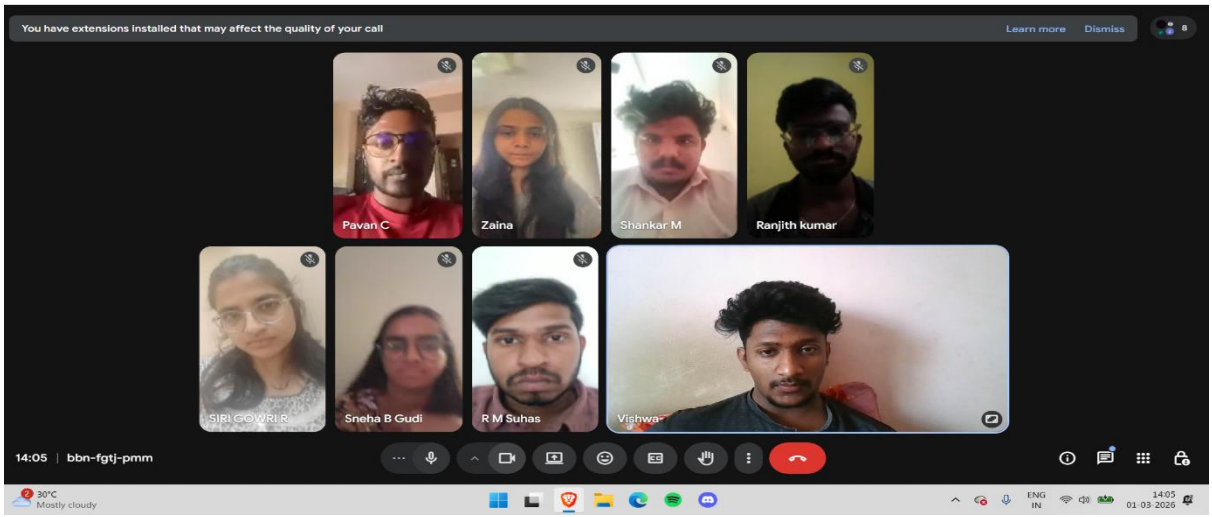


Figure: Team Meeting – 01st March 2026

Figure: Team Meeting – 02nd March 2026